Credit Card Fraud Detection Final Report

Business Problem / Objective

Credit card fraud detection is an important problem in fintech and banking because financial institutions must balance fraud prevention with minimizing false positives for legitimate customers.

This project develops an end-to-end machine learning pipeline for detecting rare fraudulent credit card transactions under severe class imbalance using behavioral transaction analytics, interpretable modeling, and ranking-based fraud prioritization.

Fraudulent transactions represented only approximately 0.58% of the dataset, making this a highly imbalanced classification problem.

Why This Project Matters

Fraud detection is a realistic machine learning problem with strong real-world business impact because it directly affects financial losses, customer trust, operational investigation costs, and customer experience.

Financial institutions must detect fraudulent activity while avoiding unnecessary blocking of legitimate customer transactions.

This project focuses not only on predictive performance, but also on business-oriented fraud prioritization, operational tradeoffs, false-positive reduction, and model interpretability.

The project also demonstrates how behavioral transaction patterns often provide stronger predictive signals than demographic information.

Dataset Overview

This project used a simulated credit card fraud detection dataset from Kaggle designed to mimic realistic transaction behavior and fraud scenarios.

The dataset is simulated rather than collected directly from live banking systems due to privacy and regulatory concerns surrounding real financial transaction data.

However, simulated fraud datasets are still commonly used for fraud detection research because they preserve realistic behavioral transaction patterns while protecting sensitive customer information.

Dataset Size

- Training dataset: 1,296,675 transactions

- Testing dataset: 555,719 transactions

- Fraud rate: ~0.58%

Key Variables

Behavioral Features

- transaction amount

- transaction speed

- transaction distance

- transaction time gaps

- merchant category

- transaction timing behavior

Temporal Features

- hour of day

- weekday

- weekend indicators

- cyclical time encoding

Demographic Features

- gender

- occupation

- city

- state

Methodology

The project followed a complete machine learning workflow:

1. Data cleaning

2. Exploratory data analysis (EDA)

3. Feature engineering

4. Leakage-safe preprocessing

5. Model training and comparison

6. Final deployment evaluation

7. SHAP interpretability analysis

Feature Engineering

Several engineered behavioral features were created, including transaction distance, transaction speed, time gaps between transactions, nighttime transaction indicators, cyclical time encoding using sine/cosine transformations, merchant normalization, log-transformed transaction amount features, and customer age features.

Hours in a day are naturally cyclical. For example, 11 PM and midnight are behaviorally close even though their numerical values appear far apart. To better capture temporal transaction behavior, cyclical time encoding using sine and cosine transformations was applied.

Some engineered behavioral features, such as transaction distance, transaction speed, and transaction time gaps, naturally interacted with each other.

Tree-based ensemble models such as XGBoost and CatBoost were well-suited for automatically capturing these nonlinear feature interactions without requiring extensive manual interaction engineering.

Special attention was also paid to minimizing target leakage through train-only preprocessing, train-only target encoding, train-only transformation artifacts, and time-aware validation procedures.

Exploratory Data Analysis (EDA)

EDA revealed several important fraud behavior patterns:

- Fraudulent transactions tended to involve larger transaction amounts.

- Certain merchant categories showed significantly higher fraud rates.

- Behavioral timing patterns appeared more predictive than demographic variables.

- Some nighttime transaction patterns behaved differently from daytime transactions.

These findings guided later feature engineering and model selection decisions.

Modeling

The project compared multiple machine learning models:

- Logistic Regression

- Random Forest

- XGBoost

- LightGBM

- CatBoost

Because fraud detection is highly imbalanced, traditional accuracy was not used as the primary evaluation metric.

Instead, evaluation focused on ROC-AUC, PR-AUC, Precision, Recall, F1 Score, and Precision@K / Recall@K.

Operational tradeoffs were also considered because fraud investigation teams can only review a limited number of flagged transactions daily.

Reducing false positives was important because incorrectly blocking legitimate customer transactions can negatively impact customer experience.

Results

Model Comparison

Model	**Mean ROC-AUC**	**Mean PR-AUC**
Logistic Regression	0.9119	0.4382
Random Forest	0.9809	0.8556
XGBoost	0.9972	0.8390
LightGBM	0.9866	0.8574
CatBoost	0.9955	0.9043

CatBoost achieved the strongest overall balance between fraud detection performance and false-positive control.

Final CatBoost Test Performance

- ROC-AUC: 0.9940

- PR-AUC: 0.8762

- Precision: 0.92

- Recall: 0.75

- F1 Score: 0.8230

Top-K Fraud Ranking Performance

- Top 100: Precision@K 1.0000 | Recall@K 0.0466

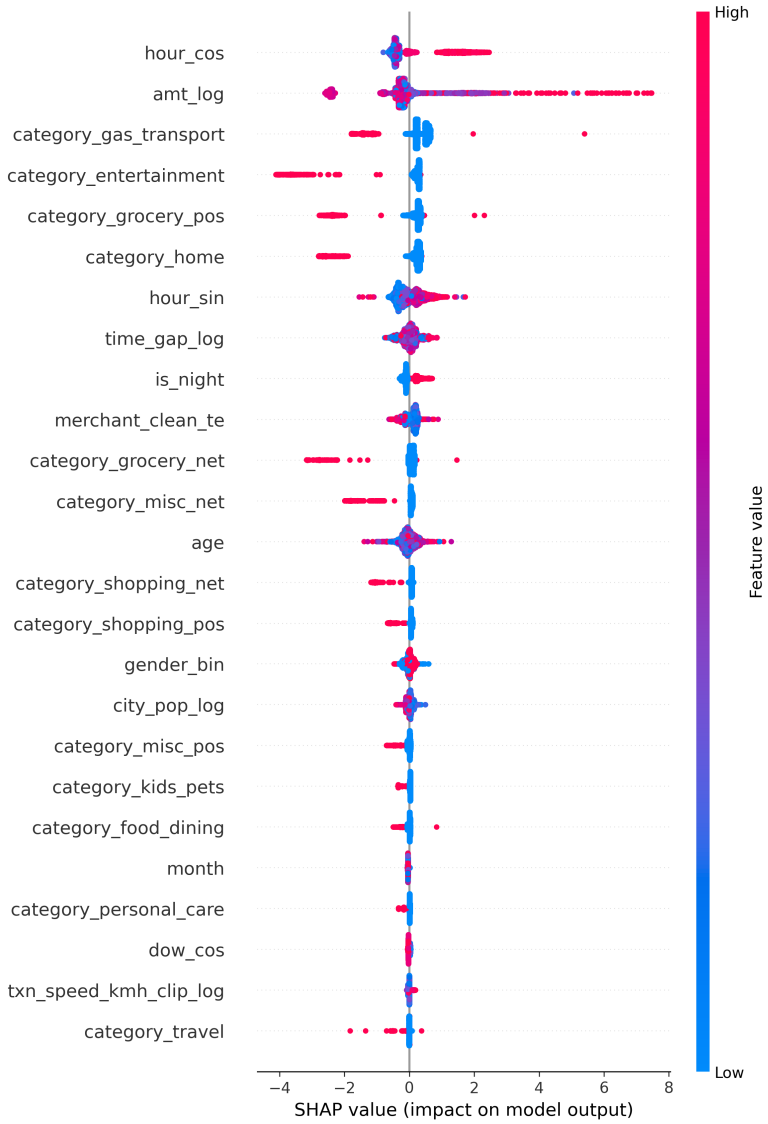
- Top 500: Precision@K 1.0000 | Recall@K 0.2331

- Top 1000: Precision@K 1.0000 | Recall@K 0.4662

- Top 5000: Precision@K 0.3914 | Recall@K 0.9124

These results demonstrate strong fraud prioritization capability for real-world fraud investigation workflows.

SHAP Interpretation



SHAP analysis was used to interpret feature importance and understand the behavioral drivers of fraud prediction.

The results showed that behavioral transaction features contributed more strongly to fraud prediction than demographic variables.

The most influential features included transaction amount, transaction timing behavior, merchant category, time gaps between transactions, and nighttime transaction behavior.

Limitations

- The dataset was simulated rather than collected directly from live banking systems.

- The project focused on offline historical transaction analysis rather than real-time fraud scoring systems.

- Fraud behavior evolves over time and may introduce concept drift in production environments.

Future Work

Potential future improvements include graph-based fraud detection, sequential deep learning models, real-time fraud scoring pipelines, fraud ring detection, drift monitoring, streaming deployment systems, enhanced explainability workflows, and threshold optimization.

Conclusion

This project demonstrated that strong fraud detection performance depends not only on model selection, but also on feature engineering, evaluation strategy, operational tradeoffs, and understanding real-world business constraints.

Behavioral transaction patterns played a particularly important role in fraud prediction.

Tree-based ensemble models significantly outperformed baseline methods under severe class imbalance, and CatBoost achieved the best overall balance between fraud detection performance and operational practicality.

Overall, this project provided valuable experience building an end-to-end machine learning workflow with applications in fintech, banking, insurance analytics, and risk modeling.